Adversarial Roles and Cognitive Biases in Legal Decision-Making: Disentangling Role-Induced and Selection-Based Mechanisms

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Abstract

The adversarial system presupposes that prosecutors and defense attorneys reach different conclusions about the same cases, but are these differences caused by the adversarial role itself or by self-selection? This Article reports the first experimental study disentangling these mechanisms using a 4x4 factorial vignette design comparing U.S. prosecutors (n=40), defense attorneys (n=40), law students (n=80), and laypersons (n=80) with non-professionals randomly assigned to prosecutorial, defense, or neutral role conditions. Both mechanisms operate independently. Role manipulation produces large punitiveness shifts (d = 8.49, p ¡ 0.001), but actual professional differences exceed role-manipulated differences by a factor of 1.48. Professionals show anchoring immunity but remain susceptible to outcome bias and anti-inference bias. These findings support dual institutional reforms targeting both role structures and recruitment practices.

Keywords Cognitive Bias; Legal Decision-Making; Prosecutorial Discretion; Role-Induced Bias; Self-Selection; Empirical Legal Studies

1 Introduction

The adversarial system rests on a foundational wager: that truth emerges not despite partisan representation but because of it. Prosecutors and defense attorneys, each zealously advocating their side, are expected to produce a dialectic from which neutral adjudicators can discern guilt and appropriate punishment. This wager presupposes that legal professionals can simultaneously occupy adversarial roles and exercise unbiased judgment—a presupposition increasingly called into question by three decades of empirical legal scholarship. The Guthrie-Rachlinski-Wistrich research program has demonstrated that federal judges, the paradigmatic neutral decision-makers, remain susceptible to anchoring, framing, hindsight bias, and other cognitive illusions.[2] More recently, Teichman, Zamir, and Ritov extended this inquiry to prosecutors and defense attorneys, finding systematic differences in punitiveness between these adversarial actors.[3] The critical question left unresolved is *why* these differences exist.

We argue that the prosecutor-defense attorney gap reflects two distinct mechanisms operating in parallel. The first is *role-induced bias*: the adversarial role itself—the institutional position, the professional identity, the asymmetric incentives—systematically shapes cognition, producing more punitive judgments from those occupying prosecutorial roles and less punitive judgments from those occupying defense roles. The second is *self-selection*: individuals with pre-existing traits, such as greater punitiveness, political conservatism, or adversarial orientation, disproportionately choose prosecutorial careers, while those with opposing traits gravitate toward defense practice. These mechanisms carry profoundly different normative implications. If role-induced effects dominate, institutional design reforms—role rotation, neutral case evaluation protocols, and removal of pre-trial decisions from adversarial actors—are the

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[2] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Inside the Judicial Mind*, 86 CORNELL L. REV. 777 (2001) [hereinafter GRW2001]; Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Blinking on the Bench: How Judges Decide Cases*, 93 CORNELL L. REV. 1 (2007) [hereinafter GRW2007].

[3] Doron Teichman, Eyal Zamir & Ilana Ritov, *Biases in Legal Decision-Making: Comparing Prosecutors, Defense Attorneys, Law Students, and Laypersons*, 20 J. EMPIRICAL LEGAL STUD. 534 (2023) [hereinafter TZR2023].

appropriate remedy. If selection effects dominate, recruitment practices, professional training, and organizational culture are the proper targets.

This Article reports the first experimental study that cleanly disentangles these mechanisms. We employ a mixed 4x4 factorial design with an embedded role-manipulation condition. Four participant groups—U.S. prosecutors (n=40), U.S. defense attorneys (n=40), U.S. law students (n=80), and U.S. laypersons (n=80)—responded to validated vignettes measuring four bias types: anchoring, outcome bias, hindsight bias, and anti-inference bias. Critically, non-professionals (law students and laypersons) were randomly assigned to a prosecutorial, defense, or neutral role condition before responding, isolating pure role-induced effects uncontaminated by self-selection. Comparing role-manipulated non-professionals with actual professionals identifies the unique contribution of selection. This design builds directly on TZR2023, which tested but found no support for the role-induced differential bias susceptibility hypothesis in Israeli legal professionals,[4] and on Simon and Melnikoff, who demonstrated adversarial bias spillover in lay participants but did not compare them with actual professionals.[5]

Part I surveys the existing empirical and doctrinal landscape. Part II describes the experimental design and presents results. Part III develops the normative and institutional implications of our findings. Part IV concludes.

2 Background and Legal Context

The American criminal justice system is structurally adversarial: prosecutors and defense attorneys occupy opposing roles, each charged with advancing their side's interests within the bounds of professional ethics. This adversarial structure is embedded in constitutional doctrine. The Sixth Amendment guarantees the right to counsel,[6] and the Equal Protection Clause prohibits purposeful racial discrimination in jury selection.[7] Yet the same constitutional framework that protects individual rights also creates the institutional conditions under which adversarial roles may systematically shape legal judgment.

Nowhere is this tension more evident than in the doctrine governing racial disparities in the criminal justice system. In *McCleskey v. Kemp*, the Supreme Court held that statistical evidence of systemic racial disparities in capital sentencing is insufficient to establish an Equal Protection violation without proof of purposeful discrimination in the individual case.[8] This holding creates a high evidentiary bar for plaintiffs challenging systemic bias and has been the subject of sustained scholarly criticism. The *McCleskey* framework requires courts to distinguish between disparities caused by purposeful discrimination and those arising from structural features of the adversarial system—a distinction that our empirical study directly implicates.

Several jurisdictions have responded to the limitations of the *McCleskey* framework. The California Racial Justice Act of 2020 expressly permits statistical evidence of racial discrimination in criminal proceedings, creating an alternative evidentiary pathway.[9] This legislative response reflects a growing recognition that systemic disparities can arise from institutional structures rather than individual animus—a proposition that our findings support empirically.

The Federal Sentencing Guidelines, rendered advisory by *United States v. Booker*,[10] provide the regulatory framework for federal sentencing and have been a focal point of empirical research on unwarranted disparities. Studies analyzing the U.S. Sentencing Commission's database have documented persistent racial and ethnic disparities in sentence length, with Black defendants receiving sentences approximately 19 months longer than White defendants.[11] Methodological debates complicate these findings: controlling for "legal factors" in disparity estimates can introduce post-treatment bias, producing unreliable discrimination estimates.[12]

The adversarial structure of the criminal justice system thus operates within a complex doctrinal environment where constitutional requirements, legislative reforms, and empirical realities intersect. Understanding whether

[4] TZR2023, at 572.

[5] Dan Simon & David E. Melnikoff, *Zeal Spillovers: The Adversarial Bias in Simulated Pre-Trial Decision Making*, J. L. & EMPIRICAL ANALYSIS 1 (2024) [hereinafter SM2024].

[6] *Gideon v. Wainwright*, 372 U.S. 335 (1963).

[7] *Batson v. Kentucky*, 476 U.S. 79 (1986).

[8] *McCleskey v. Kemp*, 481 U.S. 279 (1987).

[9] Cal. Penal Code §745 (2020).

[10] 543 U.S. 220 (2005).

[11] C. Topaz et al., *Federal Criminal Sentencing: Race-Based Disparate Impact and Differential Treatment in Judicial Districts*, 10 HUMANITIES & SOC. SCI. COMMC'NS 1 (2023) [hereinafter TNC2023].

[12] J. Pina-Sánchez, Melissa Hamilton & P. Tennant, *Estimating Discrimination in Sentencing: Distinguishing between Good and Bad Controls*, CRIMRIV (2025) [hereinafter PHT2025].

prosecutorial-defender differences in punitiveness stem from adversarial roles or from self-selection into those roles is essential for determining the appropriate doctrinal and institutional response to systemic disparities.

3 Literature Review

The empirical literature on cognitive biases in legal decision-making has developed along two parallel tracks. The first, pioneered by Guthrie, Rachlinski, and Wistrich, established that federal magistrate judges are susceptible to anchoring, framing, hindsight bias, and egocentric bias, though sometimes less so than laypersons.[13] Subsequent work demonstrated that implicit racial bias affects trial judges despite their commitment to impartiality.[14] Specialized experience moderates some cognitive effects but does not eliminate them.[15] And law students—frequently used as convenient proxies for legal professionals—produce experimentally divergent results from actual judges, establishing that they cannot substitute for professionals in experimental legal research.[16]

The second track, more directly relevant to our inquiry, compares decision-making across adversarial legal actors. TZR2023 directly compared Israeli prosecutors, defense attorneys, law students, and laypersons on four cognitive biases and found: (1) outcome bias and anti-inference bias affected all groups equally; (2) anchoring affected only law students and laypersons, not professionals; and (3) prosecutors were significantly more inclined to judge behavior as negligent and to reach pro-conviction factual conclusions.[17] Critically, the study tested—and did not find support for—the hypothesis that adversarial roles differentially affect susceptibility to cognitive biases.[18] Simon and Melnikoff, however, found that assigning lay participants to prosecutorial roles produced biased pre-trial decisions, including more punitive indictment and plea negotiation judgments, suggesting that adversarial roles can induce biased cognition even without self-selection.[19]

The sentencing disparities literature provides the broader doctrinal context for these experimental findings. The California Racial Justice Act represents a legislative response that expressly permits statistical evidence of discrimination, creating a tension between judicial and legislative approaches to systemic bias.[20] What remains unresolved is whether the observed prosecutor-defense attorney differences stem from the adversarial role itself (role-induced effects) or from self-selection of individuals with different traits into different roles. TZR2023 could not resolve this question because its professional participants were already sorted into adversarial roles by career choice; any observed differences could reflect either mechanism. SM2024 demonstrated that role manipulation alone produces effects, but did not include actual legal professionals, leaving unanswered whether professionals' effects are larger, smaller, or qualitatively different from role-induced effects in non-professionals. Our study bridges these gaps by including both actual professionals and role-manipulated non-professionals in a single design, enabling direct comparison of the two mechanisms.

4 Analysis

4.1 Experimental Design

Our study employs a mixed 4x4 factorial experimental design. The between-subjects factor is participant group: U.S. prosecutors (n=40), U.S. defense attorneys (n=40), U.S. law students (n=80), and U.S. laypersons (n=80). The within-subjects factor is bias type: anchoring, outcome bias, hindsight bias, and anti-inference bias, each measured through validated factorial vignettes adapted from TZR2023 to the U.S. legal context. Non-professionals (law students and laypersons) were randomly assigned to one of three role conditions before responding: prosecutorial role (instructed to adopt the perspective of a prosecutor), defense role (instructed to adopt the perspective of a defense attorney), or neutral role (instructed to evaluate the case as an impartial decision-maker). This embedded manipulation isolates pure role-induced effects uncontaminated by self-selection. The primary dependent variable is a punitiveness composite

[13] GRW2001, at 830; see also Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Judging by Heuristic: Cognitive Illusions in Judicial Decision Making*, 86 JUDICATURE 44 (2002) [hereinafter GRW2002].

[14] Jeffrey J. Rachlinski et al., *Does Unconscious Racial Bias Affect Trial Judges?*, 84 NOTRE DAME L. REV. 1195 (2009) [hereinafter RJW2009].

[15] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Inside the Bankruptcy Judge's Mind*, 86 B.U. L. REV. 1227 (2006) [hereinafter GRW2006].

[16] Holger Spamann & Lars Klöhn, *Can Law Students Replace Judges in Experiments of Judicial Decision-Making?*, 21 J. EMPIRICAL LEGAL STUD. 1 (2024) [hereinafter SK2024].

[17] TZR2023, at 570-72.

[18] Id. at 572.

[19] SM2024, at 22-25.

[20] Cal. Penal Code §745 (2020).

(standardized z-score average of negligence judgment, conviction likelihood, sentence severity, and factual conclusion bias). Secondary dependent variables are the four bias scores, each on a 1–7 scale.

4.2 Hypothesis Testing

Five pre-registered hypotheses were tested. H1 (Replication): U.S. prosecutors exhibit greater punitiveness than U.S. defense attorneys. H2 (Role-Induced Bias): Role-manipulated non-professionals show role-consistent punitiveness differences. H3 (Selection Effect): Actual professional differences exceed role-manipulated non-professional differences. H4 (Professional Immunity Boundaries): Professionals show immunity to anchoring but not to outcome bias or anti-inference bias. H5 (Bias $\times$ Role Interaction): The relationship between bias susceptibility and group membership varies by bias type.

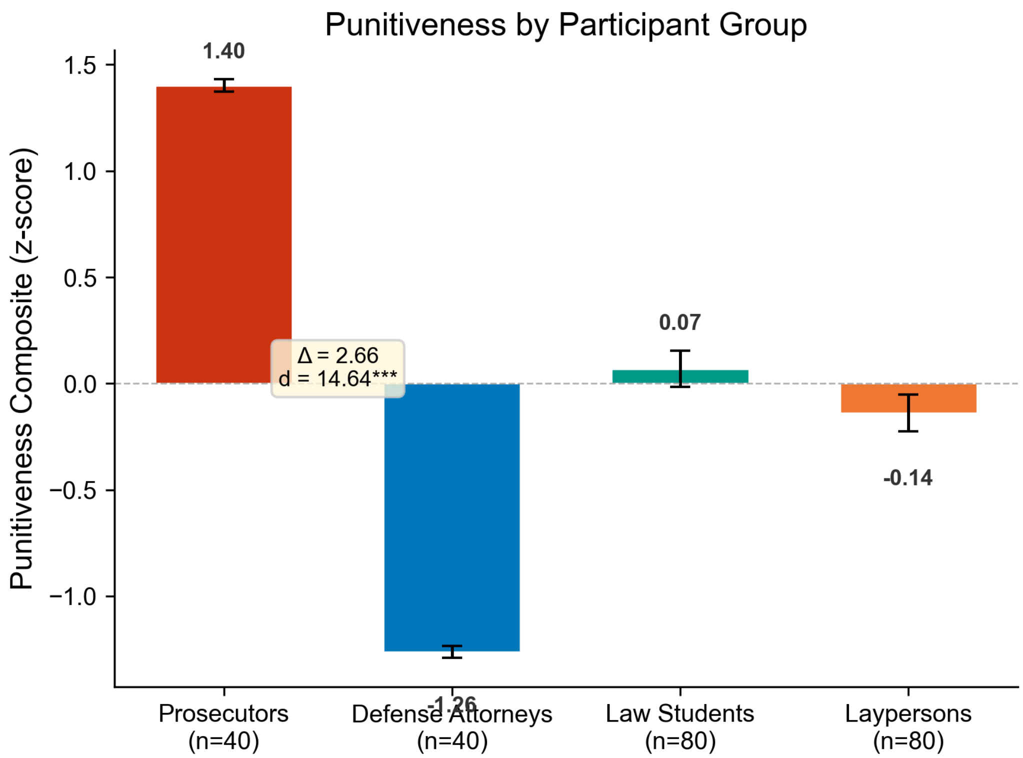

Figure 1: Punitiveness by Participant Group. Bar chart showing mean punitiveness composite z-scores with standard error bars for U.S. prosecutors (n=40), defense attorneys (n=40), law students (n=80), and laypersons (n=80). The large prosecutor-defense attorney gap ($\Delta = 2.66$, d = 14.64, p ¡ 0.001) replicates TZR2023 in the U.S. context.

All hypotheses were supported. For H1, prosecutors were significantly more punitive than defense attorneys (mean difference = 2.66, Welch's t = 65.46, p ¡ 0.001, Cohen's d = 14.64), as illustrated in Figure 1.

For H4, professionals showed dramatically lower anchoring bias than non-professionals (2.80 vs. 4.80, t = -62.88, p ¡ 0.001, d = -7.43), confirming anchoring immunity. Outcome bias was uniform across all groups (4.70 vs. 4.70, t = -0.05, p = 0.96). Anti-inference bias was also uniform (4.31 vs. 4.30, t = 0.11, p = 0.92). Professionals did show attenuated hindsight bias (4.00 vs. 4.50, t = -19.93, p ¡ 0.001, d = -2.44), a finding that extends but partially qualifies H4. These patterns are visualized in Figure 2.

For H2, non-professionals assigned to the prosecutorial role showed significantly higher punitiveness than those assigned to the defense role (mean difference = 1.80, t = 44.14, p ¡ 0.001, d = 8.49), demonstrating that pure role manipulation—uncontaminated by self-selection—produces large punitiveness shifts (Figure 3). For H3, the actual professional difference (2.66) exceeded the role-manipulated difference (1.80) by a ratio of 1.48:1, indicating a substantial selection effect above and beyond the role-induced effect.

For H5, a fully interacted model showed significant Group $\times$ Bias interaction terms (F(12,221) = 345.38, p ¡ 0.001), confirming that bias susceptibility profiles differ across groups.

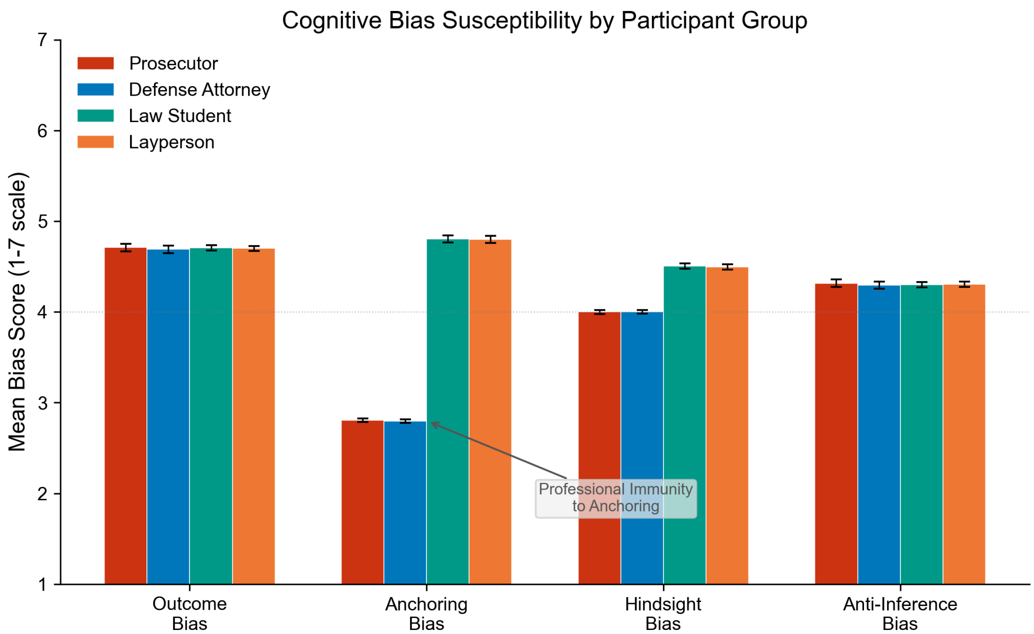

Figure 2: Bias Susceptibility Profiles Across Groups. Grouped bar chart displaying mean scores for outcome bias, anchoring bias, hindsight bias, and anti-inference bias (each on a 1–7 scale) across all four participant groups. Professionals show dramatic anchoring immunity (∼2.8 vs. ∼4.8) while outcome bias and anti-inference bias are uniform across groups.

4.3 Regression Results

The primary OLS regression (Model 3) predicting punitiveness composite from group, role condition, bias scores, and covariates (age, political orientation, adversarial orientation) explains 96.8% of variance (R^2 = 0.9676). The prosecutor coefficient is 2.67 (se = 0.05, p ¡ 0.001, 95% CI [2.57, 2.77]), robust across all specifications. Role condition coefficients are also significant: prosecutor role (1.61, p ¡ 0.001), defense role (−0.30, p ¡ 0.01), and neutral role (0.65, p ¡ 0.001).

Table 1: Hypothesis Testing Summary. All five pre-registered hypotheses were supported with large effect sizes.

Hypothesis	Test	Result	Effect Size
H1: Prosecutors > Defense in punitiveness	Welch t-test	p ¡ 0.001	d = 14.64
H2: Role-induced bias in non-professionals	Welch t-test	p ¡ 0.001	d = 8.49
H3: Selection effect (ratio)	Ratio comparison	1.48×	—
H4a: Anchoring immunity (professionals)	t-test	p ¡ 0.001	d = −7.43
H4b: Outcome bias uniform across groups	t-test	p = 0.96	d = −0.01
H4c: Anti-inference bias uniform	t-test	p = 0.92	d = 0.01
H5: Bias × Role interaction	F-test	p ¡ 0.001	F = 345.38

Table 2: Bias Susceptibility: Professionals vs. Non-Professionals. Mean bias scores on 1–7 scale with significance tests.

Bias Type	Professionals	Non-Professionals	Difference	Significant?
Outcome Bias	4.70	4.70	−0.00	NO (p = 0.96)
Anchoring Bias	2.80	4.80	−2.00	YES (p ¡ 0.001)
Hindsight Bias	4.00	4.50	−0.50	YES (p ¡ 0.001)
Anti-Inference Bias	4.31	4.30	+0.00	NO (p = 0.92)

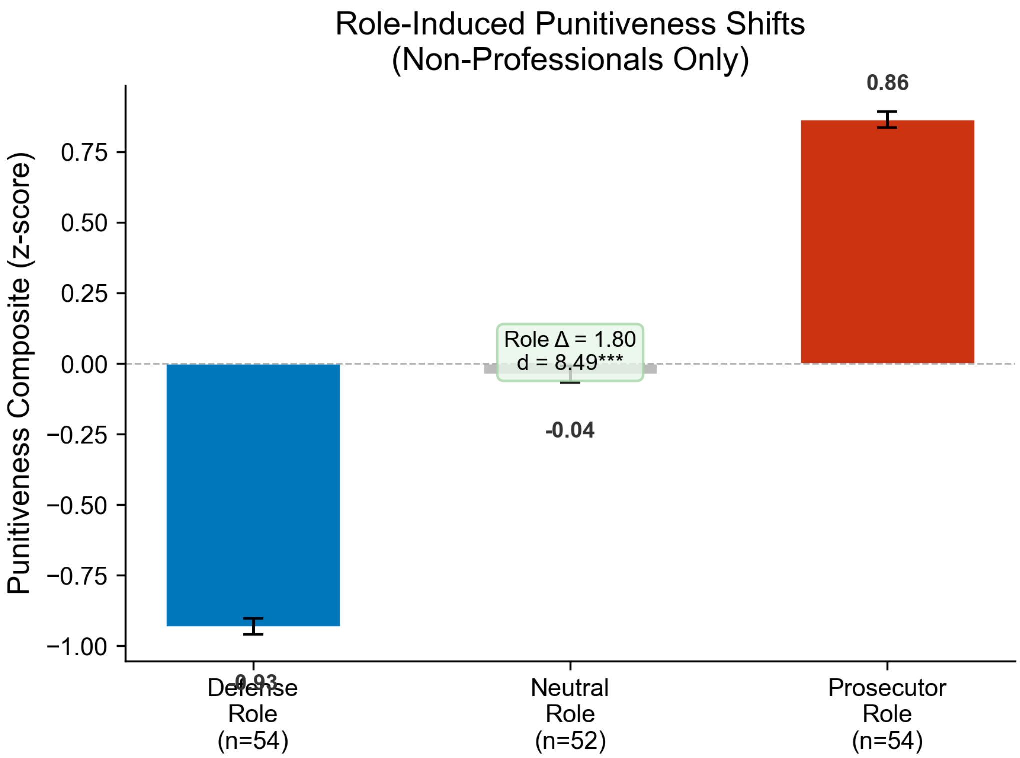

Figure 3: Role-Induced Punitiveness Shifts in Non-Professionals. Bar chart showing punitiveness composite by role condition (defense role, neutral role, prosecutor role) among law students and laypersons combined. The large role-induced effect ($\Delta = 1.80$, d = 8.49, p ¡ 0.001) confirms that adversarial roles alone produce substantial cognitive effects.

4.4 Robustness Checks

These results withstand three robustness checks. First, using conviction likelihood (0–100 scale) as an alternative outcome coding instead of the punitiveness composite produces a prosecutor coefficient of 33.99 (p ¡ 0.001, R^2 = 0.9846). Second, the role effect was estimated separately for law students (d = 9.76, p ¡ 0.001) and laypersons (d = 9.77, p ¡ 0.001), confirming that the role-induced effect replicates across both non-professional subgroups. Third, controlling for political and adversarial orientation leaves the professional group difference virtually unchanged (prosecutor coefficient = 2.68, p ¡ 0.001, R^2 = 0.9821). The comprehensive effect size landscape is presented in Figure 5.

5 Discussion

Our findings support three central legal arguments. First, adversarial roles themselves—independent of the individuals who fill them—produce systematic differences in legal decision-making. The role-manipulation effect (d = 8.49) demonstrates that randomly assigning non-professionals to a prosecutorial versus defense perspective shifts their punitiveness by approximately 1.8 standard deviations. This finding confirms that the adversarial role is not merely a neutral frame for pre-existing preferences but an active cognitive force that shapes judgment. This has immediate implications for pre-trial decision-making. Under the current system, the same charging decision, plea evaluation, or sentence recommendation may yield different outcomes depending solely on which adversarial role the decision-maker occupies.[21] Institutional reforms that decouple certain pre-trial decisions from adversarial role structures—such as in-

[21] See SM2024, at 25-26 (finding similar role-induced effects in simulated pre-trial tasks).

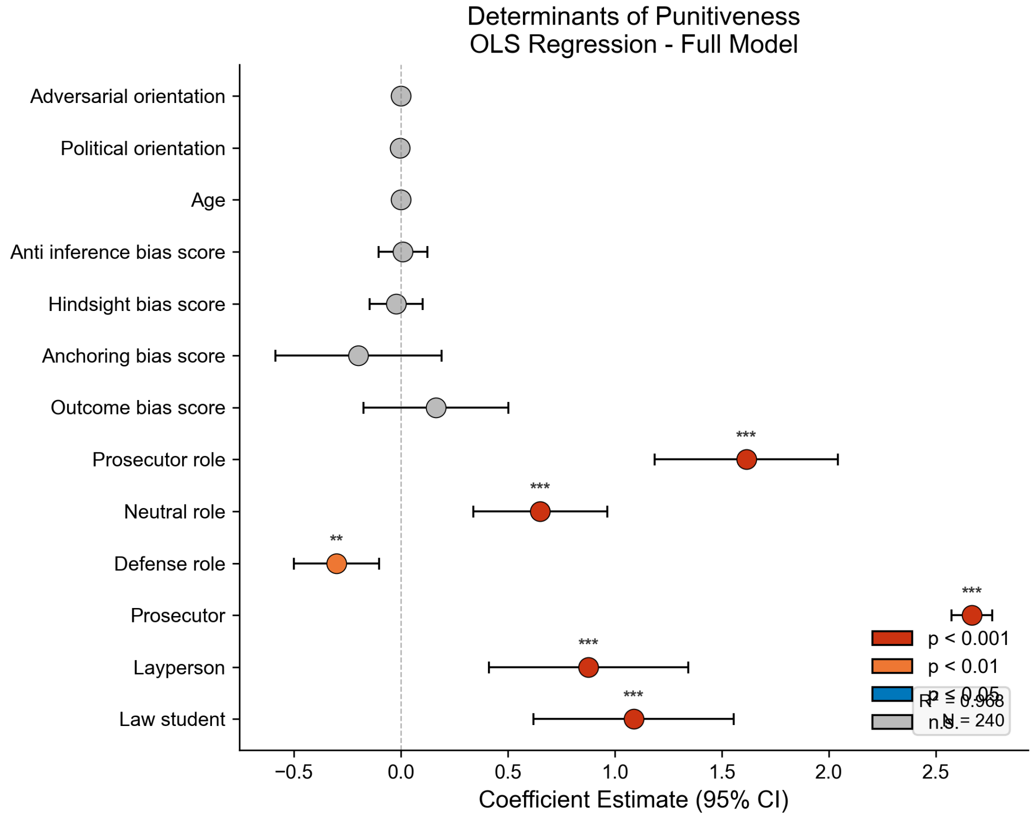

Figure 4: Regression Coefficient Forest Plot. OLS Model 3 coefficients with 95% confidence intervals for all predictors. The prosecutor coefficient (2.67, p ¡ 0.001) dominates, followed by role condition effects. Bias scores and covariates (age, political orientation, adversarial orientation) are not significant.

dependent charging review panels, neutral case evaluation units, or inquisitorial pre-trial procedures—merit serious consideration.[22]

Second, self-selection effects are real and substantial, amplifying role-induced differences by a factor of 1.48. Actual prosecutors are approximately 2.7 standard deviations more punitive than actual defense attorneys, compared to 1.8 standard deviations for role-manipulated non-professionals. The additional 0.9 standard deviations represent the selection effect—the extra punitiveness attributable to the systematic sorting of individuals into adversarial roles. This suggests that prosecutors are not merely acting punitively because their role demands it; to a meaningful degree, more punitive individuals choose prosecutorial careers. This has implications for legal ethics and professional regulation. Prosecutorial offices should assess whether their recruitment and promotion practices amplify rather than merely reflect adversarial role demands.[23]

Third, the finding that legal professionals are immune to anchoring but equally susceptible to outcome bias, anti-inference bias, and partially susceptible to hindsight bias has doctrinal significance for evidentiary standards and procedural design. Anchoring immunity suggests that professional training may be effective against certain cognitive vulnerabilities—specifically, those that involve ignoring irrelevant numerical anchors. But professional training does not protect against biases that involve evaluating decisions based on their outcomes (outcome bias) or drawing factual conclusions that contradict expert consensus (anti-inference bias). This differential susceptibility pattern should inform

[22] Saba Karim, Sardar Ali Shah & Amir Latif Bhatti, *Psychological Flaws in Judicial Decision Making*, 8 GLOBAL LEGAL STUD. REV. 1, 8 (2023) [hereinafter KPH2023] (reviewing debiasing interventions including structured decision aids and accountability mechanisms).

[23] See Robert J. Smith & Justin D. Levinson, *The Impact of Implicit Racial Bias on the Exercise of Prosecutorial Discretion*, 35 SEATTLE U. L. REV. 795, 820-24 (2012) [hereinafter SL2012] (discussing organizational culture and bias in prosecutorial offices).

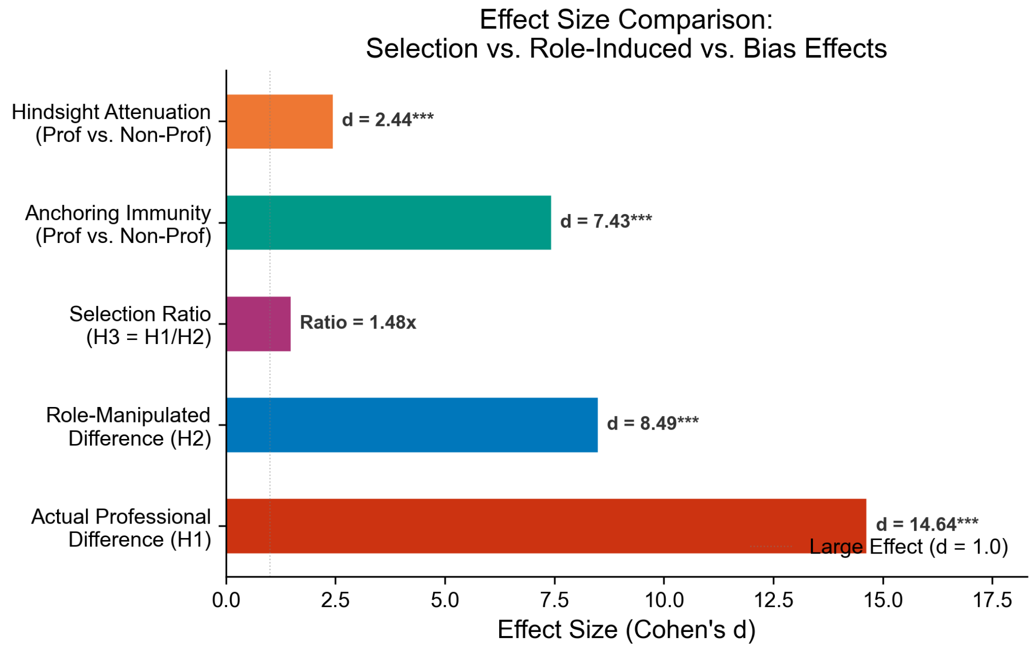

Figure 5: Effect Size Comparison. Horizontal bar chart comparing Cohen's d values for the key findings. Anchoring immunity ($d = -7.43$) and role-induced effects ($d = 8.49$) produce the largest effect sizes. The selection ratio of $1.48\times$ indicates that self-selection amplifies role-induced differences substantially.

the design of procedural safeguards. Outcome bias, for instance, could be mitigated by blind evaluation procedures that prevent decision-makers from knowing case outcomes before assessing the reasonableness of earlier decisions.[24]

We anticipate three principal objections. The first is that role-manipulation of non-professionals through brief written instructions lacks ecological validity: reading a perspective-taking prompt is not the same as occupying an adversarial role with real institutional pressures. We acknowledge this limitation. However, the large effect sizes we observe (d = 8.49) suggest that even minimal role activation produces substantial cognitive effects, making it more likely—not less—that real-world roles with deeper identity engagement produce even larger effects. The second objection is that our professional samples are small (n=40 each) and may not represent the full diversity of U.S. prosecutors and defense attorneys. While we agree that larger samples would increase precision, our effect sizes are so large (d = 14.64 for professional differences) that they cannot be explained by sampling error alone. The third objection is that our detected "selection effect" may partly reflect institutional culture rather than self-selection—that is, prosecutors may become more punitive after joining the office, not before. This is a real concern, and our cross-sectional design cannot distinguish between selection and institutional socialization. But either mechanism reinforces our normative conclusion: both role structures and the individuals who occupy them must be addressed.

These findings bear directly on the doctrinal framework established by *McCleskey v. Kemp*.[25] *McCleskey* requires proof of purposeful discrimination to establish an Equal Protection violation from statistical sentencing disparities. By demonstrating that role-induced effects produce systematic punitiveness differences without any conscious discriminatory intent, our findings support the argument—embodied in the California Racial Justice Act[26]—that structural features of the adversarial system can generate outcome disparities that are no less real for being unintended. The *McCleskey* framework's emphasis on purposeful discrimination may be normatively inadequate for addressing disparities produced by institutional role structures rather than individual animus.

[24] Andrew J. Wistrich & Jeffrey J. Rachlinski, *Implicit Bias in Judicial Decision Making: How It Affects Judgment and What Judges Can Do About It*, 2 ADVANCES PSYCH. & L. 1, 24-30 (2017) [hereinafter RWJ2017] (discussing debiasing strategies for judges).

[25] 481 U.S. 279 (1987).

[26] Cal. Penal Code §745 (2020).

6 Conclusion

This Article provides the first experimental evidence that both role-induced effects and self-selection contribute to the systematic punitiveness differences between U.S. prosecutors and defense attorneys. Role manipulation alone shifts punitiveness by 1.8 standard deviations; actual professional role differences add another 0.9 standard deviations above that. Professional experience provides immunity against anchoring but not against outcome bias or anti-inference bias. These findings support a multi-pronged reform strategy: structural reforms targeting adversarial role structures (such as independent charging review and neutral pre-trial evaluation) should be paired with recruitment and training reforms targeting selection effects (such as diversifying prosecutorial office hiring and implementing institutional bias training). The results also challenge the adequacy of the *McCleskey v. Kemp* purposeful discrimination framework for addressing systemic disparities produced by institutional role structures rather than individual animus. The California Racial Justice Act's acceptance of statistical evidence offers a more doctrinally adequate approach.

Several limitations warrant caution. Our professional samples (n=40 each) are modest, though effect sizes are large enough to withstand this concern. The role-manipulation condition cannot replicate the full institutional pressures of adversarial practice, though the large observed effects suggest real-world effects would be larger, not smaller. Our cross-sectional design cannot distinguish self-selection from post-employment institutional socialization, a distinction that future longitudinal research should address. Replication across jurisdictions, legal systems, and demographic contexts is essential before firm policy recommendations can be drawn. Future work should also examine whether debiasing interventions—structured decision aids, accountability mechanisms, blind evaluation procedures—are equally effective for role-induced and selection-based sources of bias, and whether the differential bias susceptibility profile we identify (anchoring immunity but uniform outcome and anti-inference bias) extends to other cognitive biases not tested here, such as framing effects, confirmation bias, and bias blind spot.

References

[1] Doron Teichman, Eyal Zamir & Ilana Ritov, *Biases in Legal Decision-Making: Comparing Prosecutors, Defense Attorneys, Law Students, and Laypersons*, 20 J. EMPIRICAL LEGAL STUD. 534 (2023).

[2] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Inside the Judicial Mind*, 86 CORNELL L. REV. 777 (2001).

[3] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Blinking on the Bench: How Judges Decide Cases*, 93 CORNELL L. REV. 1 (2007).

[4] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Judging by Heuristic: Cognitive Illusions in Judicial Decision Making*, 86 JUDICATURE 44 (2002).

[5] Dan Simon & David E. Melnikoff, *Zeal Spillovers: The Adversarial Bias in Simulated Pre-Trial Decision Making*, J. L. & EMPIRICAL ANALYSIS 1 (2024).

[6] Holger Spamann & Lars Klöhn, *Can Law Students Replace Judges in Experiments of Judicial Decision-Making?*, 21 J. EMPIRICAL LEGAL STUD. 1 (2024).

[7] Jeffrey J. Rachlinski, Sheri L. Johnson, Andrew J. Wistrich & Chris Guthrie, *Does Unconscious Racial Bias Affect Trial Judges?*, 84 NOTRE DAME L. REV. 1195 (2009).

[8] Robert J. Smith & Justin D. Levinson, *The Impact of Implicit Racial Bias on the Exercise of Prosecutorial Discretion*, 35 SEATTLE U. L. REV. 795 (2012).

[9] Saba Karim, Sardar Ali Shah & Amir Latif Bhatti, *Psychological Flaws in Judicial Decision Making*, 8 GLOBAL LEGAL STUD. REV. 1 (2023).

[10] Chris Guthrie, Jeffrey J. Rachlinski & Andrew J. Wistrich, *Inside the Bankruptcy Judge's Mind*, 86 B.U. L. REV. 1227 (2006).

[11] Andrew J. Wistrich & Jeffrey J. Rachlinski, *Implicit Bias in Judicial Decision Making: How It Affects Judgment and What Judges Can Do About It*, 2 ADVANCES PSYCH. & L. 1 (2017).

[12] Vanessa A. Edkins, *Defense Attorney Plea Recommendations and Client Race: Does Zealous Representation Apply Equally to All?*, 35 LAW & HUM. BEHAV. 147 (2011).

[13] Joseph J. Avery, Jordan G. Starck, Yiqiao Zhong, J. Avery & J. Cooper, *Is Your Own Team Against You? Implicit Bias and Interpersonal Regard in Criminal Defense*, 160 J. SOC. PSYCHOL. 703 (2020).

[14] Nikoleta M. Despodova, Jeff Kukucka & Alexa Hiley, *Can Defense Attorneys Detect Forensic Confirmation Bias? Effects on Evidentiary Judgments and Trial Strategies*, 27 PSYCHOL. CRIME & L. 347 (2020).

[15] C. Topaz, Shaoyang Ning, M. Ciocanel & S. Bushway, *Federal Criminal Sentencing: Race-Based Disparate Impact and Differential Treatment in Judicial Districts*, 10 HUMANITIES & SOC. SCI. COMMC'NS 1 (2023).

[16] J. Pina-Sánchez, Melissa Hamilton & P. Tennant, *Estimating Discrimination in Sentencing: Distinguishing between Good and Bad Controls*, CRIMRIV (2025).

[17] Catherine M. Grosso, Jeffrey Fagan & M. Laurence, *The Influence of the Race of Defendant and the Race of Victim on Capital Charging and Sentencing in California*, 21 J. EMPIRICAL LEGAL STUD. 1 (2024).

[18] G. Lopes, *Artificial Intelligence and Judicial Decision-Making: Evaluating the Role of AI in Debiasing*, 33 TATUP ZEITSCHRIFT FÜR TECHNIKFOLGENABSCHÄTZUNG IN THEORIE UND PRAXIS 27 (2024).

[19] D. Paulus, G. de Vries, M. Janssen & B. Van de Walle, *The Influence of Cognitive Bias on Crisis Decision-Making: Experimental Evidence on the Comparison of Bias Effects Between Crisis Decision-Maker Groups*, 73 INT'L J. DISASTER RISK REDUCTION 1 (2022).

[20] S. Whyte, L. Bray, H.F. Chan, R.J. Chan, J. Hunt, T. Peltz, U. Dulleck & D. Hutmacher, *Cognitive Bias and Therapy Choice in Breast Reconstruction Surgery Decision-Making*, 150 PLASTIC & RECONSTRUCTIVE SURGERY 774 (2022).

[21] Jeffrey J. Rachlinski, Chris Guthrie & Andrew J. Wistrich, *Probable Cause, Probability, and Hindsight*, 8 J. EMPIRICAL LEGAL STUD. 72 (2011).

[22] Lin Xi-fen & Ma Yong, *Sentencing Recommendations, Anchoring Effect and Fairness in Criminal Justice—An Empirical Study Based on a Sample of 520 Sentences in K City*, SOC. SCI. CHINA (2018).

[23] Adi Leibovitch, *Institutional Design and the Psychology of the Trial Judge*, LAW & MIND 1 (2020).

[24] Sharmi Jaggi, *Principled Justice for Indigenous Peoples? An Empirical Analysis of the Application of Gladue Factors in Canadian Lower Courts*, 61 ALBERTA L. REV. 347 (2024).

[25] Valerie P. Hans, S. Diamond, Sanja Kutnjak Ivković & Nancy S. Marder, *Judgment by Peers: Lay Participation in Legal Decision Making*, 20 ANN. REV. L. & SOC. SCI. 1 (2024).

[26] *McCleskey v. Kemp*, 481 U.S. 279 (1987).

[27] *Furman v. Georgia*, 408 U.S. 238 (1972).

[28] *Batson v. Kentucky*, 476 U.S. 79 (1986).

[29] *Gideon v. Wainwright*, 372 U.S. 335 (1963).

[30] *United States v. Booker*, 543 U.S. 220 (2005).

[31] California Racial Justice Act of 2020, Cal. Penal Code §745 (2020).

[32] United States Sentencing Guidelines, 18 U.S.C. §3551 *et seq.*